

Integrated Photonic RF Receiver with Frequency Downconversion in a Packaged InP PIC for Satellite Payloads

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Abstract We demonstrate an integrated photonic RF front-end for satellite payloads using a packaged InP PIC, combining optical LO distribution with Ka- to C-band microwave downconversion, supporting 1×128 antenna elements, achieving 24 dB conversion loss, 53 dB noise figure, and $< 15\%$ EVM across modulation formats. ©2026 The Author(s)

Introduction

Photonic technologies are increasingly being investigated for RF front-ends (FEs) in high-capacity satellite payloads, driven by their potential for reduced size, weight, and power (SWaP) consumption, as well as the low-loss distribution of high-frequency signals [1–3]. In particular, optical LO distribution enables low-loss routing across large arrays [4], while fiber-based links can reduce RF harness mass by up to an order of magnitude, significantly lowering payload SWaP [5]. Combined with photonic integrated circuits (PICs), this enables compact and scalable architectures suitable for mm-wave operation and dense phased-array antennas while supporting functionalities such as optical beamforming and photonic signal processing [3, 6, 7].

The concept of optical LO distribution for satcom applications has been investigated [4, 8], and integrated photonic RF receivers have shown promising front-end performance [6, 9–11]. However, these approaches have not been demonstrated in architectures targeting phased-array antennas (PAAs) with a large number of radiating elements. A complete integrated photonic RF FE, combining optical LO distribution with frequency downconversion to an intermediate frequency (IF), is required for practical receiver architectures; yet, such a system has not been demonstrated. In our previous work [12], we characterized the devices on a packaged InP PIC, establishing their RF performance as a step toward this architecture.

In this paper, we demonstrate an integrated photonic RF FE system that combines optical LO distribution with optical microwave downconversion from the Ka-band (27.5 GHz to 30.5 GHz) to the C-band (3.9 GHz to 6.9 GHz) in the packaged InP PIC [12]. The C-band corresponds to the input frequency range of digital transparent processors (DTPs) in satellite payloads [13]. The RF performance is characterized in terms of conversion gain and noise figure. In addition, system-level mea-

surements show an error vector magnitude below 15% across multiple modulation formats for optical LO distribution to 1×128 antenna elements. The results establish that integrated photonic RF front-ends are feasible for real satellite applications, offering reduced SWaP with performance comparable to conventional electrical RF FEs.

System Architecture and Implementation

The photonic RF FE system is shown in Fig. 1 for local oscillator (LO) signal distribution to the antenna elements of a PAA. The antenna system is typically at the payload edge or exterior, while digital signal processing units, such as the digital transparent processor (DTP), are located in the payload core. A continuous-wave laser is modulated by the external LO source using a null-biased Mach–Zehnder modulator (MZM) to generate a carrier-suppressed double-sideband (CS-DSB) signal, enabling optical frequency doubling and effectively halving the required LO source frequency f_{LO} . To compensate for MZM insertion loss, the signal is amplified by a high-power optical amplifier (HPOA) and attenuated to emulate high splitting ratios in large antenna arrays. The received Ka-band RF signal is applied to a phase modulator (PM), resulting in an optical field carrying both LO and RF signals.

At the receiver, the optical signal is first amplified using a semiconductor optical amplifier (SOA) to compensate for coupling and on-chip waveguide losses. The phase-modulated signal is then converted to intensity modulation by a Mach–Zehnder interferometer (MZI), biased at quadrature and supporting broadband RF operation from Ka-band (27.5 GHz to 30.5 GHz) to V-band (47.2 GHz to 50.2 GHz) [12]. The two complementary outputs of the MZI are detected using a balanced photodetector (BPD), enabling improved noise performance via common-mode suppression. Through heterodyne beating of the optical LO and RF tones, the signal is downconverted to an IF in the C-band (3.9 GHz to 6.9 GHz). The IF signals are further

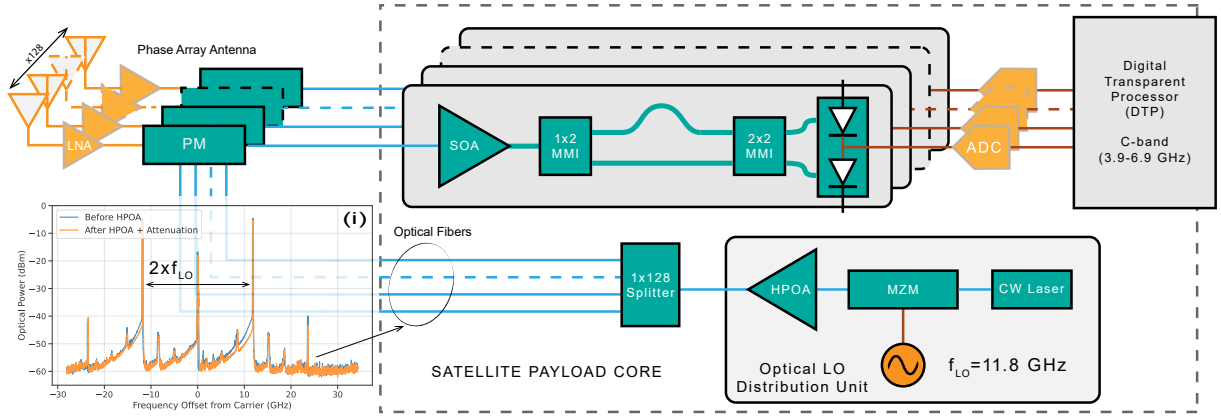


Fig. 1: Integrated photonics RF receiver FE for satellite payloads. The architecture includes an optical LO signal distribution unit, phase modulators (PMs) co-located with the phased-array antenna, and a centralized photonic receiver subsystem. Both the PMs and receiver are integrated on a PIC; LO distribution in the experiment was emulated using an optical attenuator to mimic splitter losses. Inset (i) shows the optical LO spectrum distributed to each antenna element.

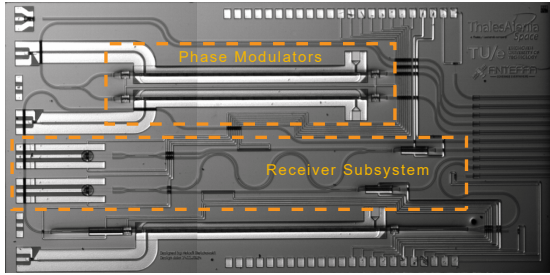


Fig. 2: InP PIC featuring phase modulators and a receiver subsystem with an SOA, MZI, and BPD. The chip packaging is presented in [12].

sampled and processed by a digital transparent processor (DTP).

This approach is particularly beneficial for large satellite payloads, such as geostationary satellites with hundreds of antenna elements. In such scenarios, the use of phase modulators removes the need for bias controllers, significantly simplifying system operation, reducing the number of control signals, and improving the optical power budget compared to using MZMs at the antenna site. In this work, the phase modulator and receiver subsystem are implemented on the same packaged InP PIC, with separate optical inputs and outputs to emulate a practical RF FE configuration in which the transmitter and receiver sub-systems are physically separated. The PIC implementation is shown in Fig. 2, more details on the packaging can be found in [12].

Results and Discussion

The external LO at $f_{LO} = 11.8$ GHz drives the null-biased MZM at a modulation index of $m = \frac{\pi V_{RF}}{V_{\pi}} = 0.6$, limited by the available RF power. This sub-optimal drive reduces the efficiency of optical LO generation by limiting the power in the first-order sidebands (see Fig. 1), resulting in an optical input power of 4.0 dBm to the HPOA. Driving the MZM at the optimum $m = 1.84$ would increase the combined sideband power by approximately 6.1 dB, thereby improving the optical SNR at the HPOA input. With the HPOA output fixed at 24 dBm, the

reduced input SNR requires higher gain, leading to increased excess RIN [14]. Increasing the MZM drive power would therefore mitigate this excess RIN. A high MZM extinction ratio (> 25 dB) is also critical, as residual carrier leakage is amplified and can appear with significant power in the output electrical spectrum (see Fig. 1(i)).

The RF performance was characterized in terms of conversion gain G_{IF} and noise figure NF for a signal at 29 GHz downconverted to 5 GHz, as shown in Fig. 3. The optical input power to the PIC was swept from -15 dBm to 10 dBm, emulating different LO distribution scenarios corresponding to varying splitting ratios and, consequently, different numbers of fed antenna elements, ranging from approximately 1×16 to 1×1024 . The results are presented as a function of the total BPD DC current I_{DC} , which can be measured directly since the on-chip optical power at the BPD is not accessible. The corresponding optical power can be estimated using a measured single-photodiode responsivity of 0.46 A/W.

As the BPD current increases, the G_{IF} rises rapidly at low currents and begins to saturate at moderate levels around 4 mA, reaching a maximum of approximately -24 dB—comparable to conventional electrical RF front-ends. On a per-photodiode basis, saturation occurs slightly above 2 mA. This behavior is not limited to the photodiodes themselves, as the stand-alone PD shows saturation currents exceeding 10 mA. Instead, the observed compression is most likely caused by SOA saturation at higher input optical powers. The relatively low G_{IF} is further influenced by the RF phase modulator, which has a half-wave voltage of $V_{\pi} = 7$ V at $f_{RF} = 29$ GHz. Since the conversion efficiency scales inversely with V_{π}^2 , this directly limits the achievable gain [2]. Strategies to reduce V_{π} and improve the modulation efficiency of the packaged InP PIC were discussed in [12].

The NF is plotted versus the total BPD DC current I_{DC} in Fig. 3. The HPOA and SOA

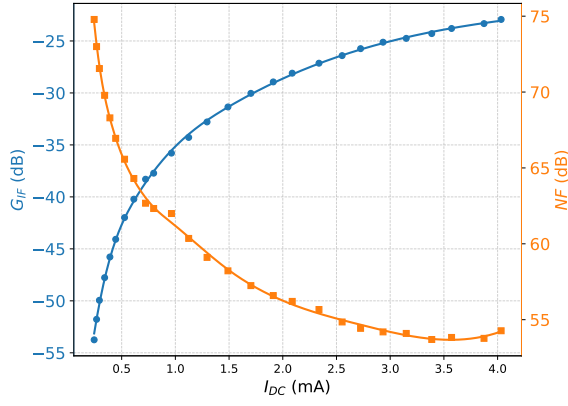


Fig. 3: The conversion gain G_{IF} and the noise figure NF as a function of the total BPD DC current at $f_{IF} = 5$ GHz.

have reported NF s of 4 dB and 7 dB, respectively. The minimum measured NF is 53 dB. The NF reaches a quasi-constant value for I_{DC} higher than 3 mA, most likely due to RF gain saturation at higher optical input powers to the SOA. In this regime, both the noise PSD and signal power increase proportionally, resulting in a constant SNR. This suggests that RF performance can be improved by either removing the on-chip SOA and increasing HPOA output power or by using a higher-gain SOA. It is important to note that in practical receiver chains, photonic links are typically preceded by a high-gain, low-noise amplifier (LNA) [15], which largely determines the overall system NF . Therefore, while the absolute NF reported here is high, its impact on the end-to-end receiver performance is mitigated in realistic system deployments.

A data transmission setup was implemented to validate the constellation integrity through E/O conversion, IF downconversion, and balanced detection. Modulated signals were generated at 29 GHz using a vector signal generator (VSG) and analyzed using a vector signal analyzer (VSA), which captured the downconverted signal at 5 GHz. Modulation formats suitable for satellite communication [16] were transmitted, namely QPSK, 16-QAM, 32-QAM, 16-APSK, and 32-APSK, at a symbol rate of 23.3 MSymb/s, limited by the VSA bandwidth of approximately 28 MHz, using a root-raised cosine filter with a roll-off factor of 0.2.

The optical input power to the receiver PIC was set to -0.5 dBm, emulating a 1×128 distribution to antenna elements in a phased-array antenna. The semiconductor optical amplifier (SOA) was biased at 130 mA, providing approximately 10 dB of gain. The total current at the output of the BPD was constant at 2.2 mA, which, assuming a PD responsivity of 0.46 A/W, corresponds to a total received optical power of 6.8 dBm.

System performance was assessed via error vector magnitude (EVM), with results shown in Fig. 4. EVM remains below 15% across all formats for input RF powers between -20 dBm to

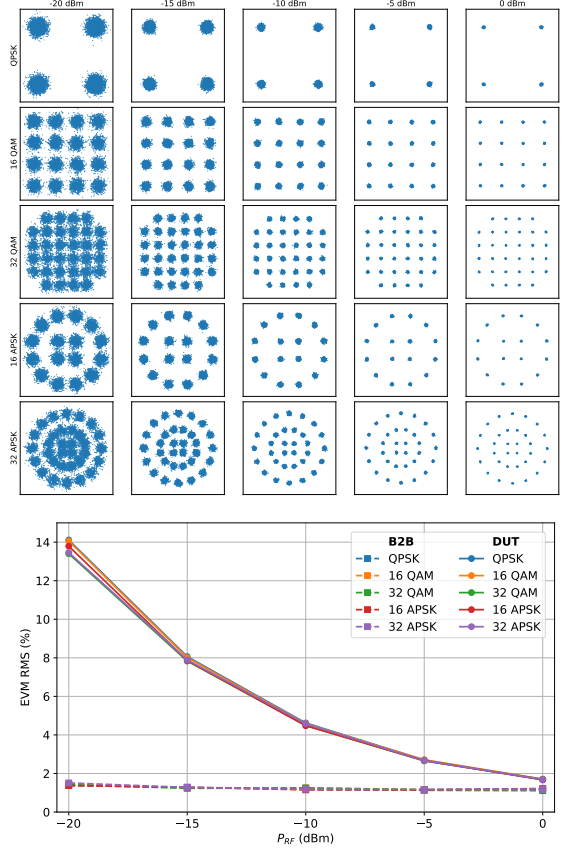


Fig. 4: (upper) Constellation evolution for selected modulation formats vs. input RF power. **(lower)** Corresponding EVM vs. input RF power for back-to-back (B2B) and device-under-test (DUT) measurements.

0 dBm, consistent with typical LNA output levels in the receiver FE. As the input RF power decreases, EVM degrades with the same trends across all formats, indicating noise-limited performance. This is attributed to the high NF of the link, mainly due to SOA-induced gain saturation, as also seen in the gain and NF behavior in Fig. 3. Nonlinearities and phase noise are not dominant in this regime. The EVM floor of approximately 2% is limited by the back-to-back (B2B) performance of the data transmission system.

Conclusions

We have demonstrated an integrated photonic RF front-end based on a packaged InP PIC, enabling optical LO distribution with simultaneous Ka-to-C-band downconversion for satellite applications. Optical fiber links replace coaxial interconnects, while phase modulators and balanced photodetectors are integrated on-chip, supporting bias-free and scalable operation. Functionality is validated by emulating the LO distribution to the 128 element antenna front-end and achieving an EVM below 15%. Although further optimization—particularly of the SOA and LO distribution—can significantly improve RF performance, the results confirm the feasibility of integrated photonic RF front-ends for high-capacity satellites.

Acknowledgments

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